FEVER: the result tables of DMS-BTT, from our data

Tables 1-5 follow DMS-BTT (Tang et al., ESWA 2025), Tables 2-6, one for one; Table 6 adds the harm to users that DMS-BTT does not report. Testbed: Mininet + Open vSwitch + Ryu, tree topology (core switch s11), table 3,500 entries, idle timeout 10 s, 300 s runs, attack from 50 s, benign traffic the last 60 % of bigFlows (T1B), which no model or threshold was trained on. 7 November 2026.

Table 1. LFTO attack parameter setting (cf. DMS-BTT Table 2).

Attack	T1 (s)	S (pkt/T)	M (pkt)	Packets per refresh	Packet size	Source	T2 (s)
A2	[5.5, 9]	100	1700	1	64 B	own address	83.5
A4	[6, 9.5]	100	1700	1-3	benign-like	address pool	103.0
E9P	[2, 3]	100	1700	1-3	benign-like	address pool	31.0

T1: sending period; S: step (new flows per burst); M: maximum attack flows per attacker (two attackers); T2: mean time from the start of the attack to the first overflow of the core table with no defense (5 seeds). A2 is DMS-BTT's parameter set 2. A4 refreshes with benign-like packets from an address pool (mimicry); E9P is the defense-aware best response found by an offline search.

Table 2. Attack detection by FEVER on the testbed, per run (cf. DMS-BTT Table 3).

Traffic	Runs	Attack detected (both attackers blocked)	Attackers blocked	Time to first block (s after attack start)	Runs with a benign host blocked
A2	5	5 / 5	10 / 10	70.7	0 / 5
A4	5	5 / 5	10 / 10	126.4	0 / 5
E9P	5	5 / 5	10 / 10	70.0	0 / 5
no attack	5	-	-	-	0 / 5

DMS-BTT classifies the network state (normal / attack / overflow) with CatBoost on FN and APIT. FEVER uses the occupancy thresholds TH1 = 70 % and TH2 = 90 % instead and detects an attack by acting on its hosts, so its detection is reported per run. v5 campaign, seeds 4140-4144.

Table 3. Results of malicious flow identification of LFTO attacks with different parameters (cf. DMS-BTT Table 4).

Attack	Accuracy	Recall	Precision	F1 Score
A2	99.39 %	99.28 %	99.95 %	99.61 %
A4	21.18 %	0.01 %	14.29 %	0.02 %
E9P	10.98 %	0.00 %	0.00 %	0.00 %
Average	43.85 %	33.10 %	38.08 %	33.21 %
Mix	42.69 %	30.66 %	99.88 %	46.92 %
Host level, testbed	**100.00 %**	**100.00 %**	**100.00 %**	**100.00 %**

Rows A2-Mix: FEVER's detector alone (the three shared CatBoost models, $\tau = 0.620$), per flow entry, on simulator T1B traffic at the testbed's rate (development seeds 2000-2002; 5 % of entries per poll). It identifies A2 almost perfectly but not A4 or E9P, whose entries look benign; this is why FEVER adds the volume block. Last row: FEVER as deployed, per host on the testbed (v5 campaign): 30 of 30 attacker hosts blocked, 0 of 80 benign hosts blocked.

Table 4. Comparing the outcomes of various classifiers to recognize harmful flows (cf. DMS-BTT Table 5).

Detector	Accuracy	Recall	Precision	F1	Recall, seen only by another controller	Recall, seen by none	F1, seen by none
FTOP (Random Forest)	96.94 %	92.02 %	99.79 %	95.02 %	79.43 %	9.49 %	14.74 %
FTMaster (XGBoost)	97.07 %	92.37 %	99.79 %	95.30 %	81.29 %	15.00 %	20.65 %
FloRa (CatBoost)	99.01 %	97.50 %	99.82 %	98.62 %	82.19 %	24.90 %	28.46 %
DMS-BTT (CatBoost)	97.03 %	92.27 %	99.78 %	95.22 %	81.62 %	14.28 %	19.94 %
FL-DualGuard (MLP, federated)	98.98 %	97.37 %	99.82 %	98.53 %	96.70 %	34.41 %	50.56 %
EnsGuard (RF + CatBoost, federated)	**99.60 %**	**99.17 %**	**99.75 %**	**99.45 %**	**98.69 %**	**48.47 %**	**64.50 %**

E11 benchmark (simulator domains D1-D3): every detector trained and scored on identical rows, each at a 0.1 % benign false-positive rate. Columns 2-5: attacks the controller trained on (DMS-BTT's setting). FEVER's detector is the CatBoost half of EnsGuard; on validation it matches the ensemble on attacks not trained on (92.5 % against 92.2 % recall, results/v4lite/detsel.md).

Table 5. Comparative experiment results (cf. DMS-BTT Table 6).

Method	Average Pa	Max Pa	Average flow table usage	Earliest overflow time (s)	Packet-In messages	Table-Full messages
None	62.10 %	94.43 %	76.58 %	21 (15/15)	101,854	3,217
EnsGuard v3	41.74 %	84.57 %	56.05 %	70 (11/15)	87,072	209
FEVER without volume (v4-lite)	43.40 %	82.52 %	59.25 %	78 (11/15)	94,241	267
DMS-BTT	41.85 %	78.24 %	58.06 %	40 (10/15)	105,310	271
FTMaster	45.54 %	82.16 %	60.88 %	31 (10/15)	111,955	1,509
FEVER	**21.94 %**	**82.86 %**	**40.32 %**	**110 (7/15)**	**118,529**	**250**

Mean over A2, A4 and E9P x 5 seeds (15 runs per method), core switch s11. Earliest overflow: seconds from the start of the attack to the table's first overflow, the earliest over the runs, with the number of runs that overflowed at all. Packet-In and Table-Full: mean per 300 s run. All methods run on the controller. DMS-BTT and FTMaster are trained on T1A. EVICT, SIFT, TableGuard and FTOP ran only in the earlier campaign (other traffic) and are not listed.

Table 5a. As Table 5, attack A2 only (5 seeds).

Method	Average Pa	Max Pa	Average flow table usage	Earliest overflow time (s)	Packet-In messages	Table-Full messages
None	58.43 %	94.79 %	75.20 %	75 (5/5)	106,937	3,350
EnsGuard v3	12.18 %	72.67 %	31.03 %	112 (3/5)	115,958	248
FEVER without volume (v4-lite)	11.88 %	65.72 %	31.93 %	115 (2/5)	113,658	96
DMS-BTT	18.67 %	54.28 %	38.55 %	never (0/5)	132,215	0
FTMaster	9.98 %	58.10 %	30.21 %	never (0/5)	158,455	0
FEVER	**12.62 %**	**84.74 %**	**31.72 %**	**110 (4/5)**	**113,602**	**622**

Table 5b. As Table 5, attack A4 only (5 seeds).

Method	Average Pa	Max Pa	Average flow table usage	Earliest overflow time (s)	Packet-In messages	Table-Full messages
None	54.77 %	91.38 %	70.94 %	83 (5/5)	99,238	1,828
EnsGuard v3	46.98 %	87.32 %	61.28 %	128 (5/5)	82,363	179
FEVER without volume (v4-lite)	50.88 %	87.08 %	67.39 %	113 (4/5)	91,714	190
DMS-BTT	47.58 %	86.21 %	64.63 %	132 (5/5)	96,989	245
FTMaster	53.36 %	91.42 %	69.06 %	104 (5/5)	90,553	1,307
FEVER	**27.23 %**	**80.04 %**	**47.69 %**	**208 (2/5)**	**118,623**	**103**

Table 5c. As Table 5, attack E9P only (5 seeds).

Method	Average Pa	Max Pa	Average flow table usage	Earliest overflow time (s)	Packet-In messages	Table-Full messages
None	73.10 %	97.13 %	83.60 %	21 (5/5)	99,386	4,472
EnsGuard v3	66.06 %	93.72 %	75.82 %	70 (3/5)	62,895	199
FEVER without volume (v4-lite)	67.43 %	94.78 %	78.43 %	78 (5/5)	77,353	516
DMS-BTT	59.30 %	94.23 %	71.00 %	40 (5/5)	86,725	568
FTMaster	73.29 %	96.97 %	83.38 %	31 (5/5)	86,856	3,219
FEVER	**25.96 %**	**83.80 %**	**41.56 %**	**244 (1/5)**	**123,360**	**26**

Table 6. Harm to users and actions taken (not reported by DMS-BTT).

Method	Benign flows harmed	Lost first packets	New connections succeeding	Decision time per poll (ms, median)	Attackers blocked	Benign hosts blocked (20 runs)
None	10.13 %	28.2 %	25.2 %	30	0 / 30	0
EnsGuard v3	3.48 %	34.3 %	27.4 %	123	10 / 30	6
FEVER without volume (v4-lite)	1.60 %	30.5 %	29.7 %	45	10 / 30	0
DMS-BTT	1.71 %	32.6 %	23.4 %	65	0 / 30	0
FTMaster	4.80 %	27.8 %	23.6 %	72	0 / 30	0
FEVER	**0.49 %**	**19.8 %**	**29.8 %**	**13**	**30 / 30**	**0**

Same 15 attack runs per method (benign hosts blocked: also the 5 no-attack runs). Benign flows harmed: refused, dropped by a block, or first packet lost. New connections: a probe opens one TCP connection every 0.5 s after the attack starts. Preregistered tests on these runs: results/testbed_prereg_v5/report.md.